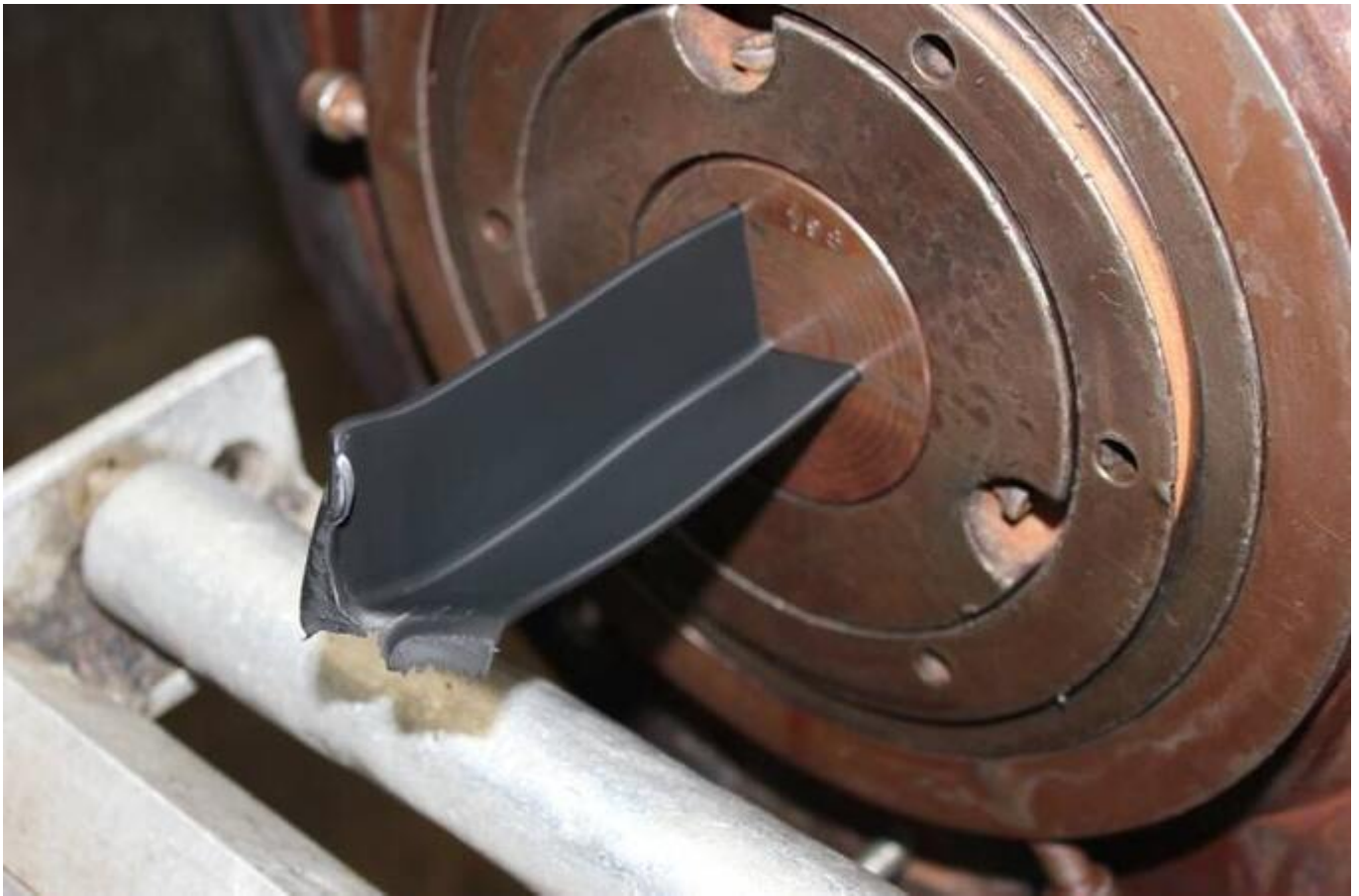


Qualification Pack



Rubber Extruder Operator

QP Code: RSC/Q2601

Version: 4.0

NSQF Level: 4

Rubber, Chemical & Petrochemical Skill development Council || 217, 2nd floor, Rectangle One, Saket
District Centre
New Delhi - 110017 || email: vishaka.v@rcpsdc.in

Qualification Pack

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Qualification Pack

RSC/Q2601: Rubber Extruder Operator

Brief Job Description

The Rubber Extruder operator sets up, operates, and maintains automated rubber extrusion machines to produce consistent rubber profiles. The Operator controls process parameters, inspects output quality, troubleshoots operational issues, and collaborates with production teams to ensure efficient and safe operations.

Personal Attributes

This job requires the individual to work independently or under supervision. He should be motivated and have a positive attitude. He should be comfortable in performing laborious work and operate machines efficiently. The individual must be result oriented and be able to work in factory environment.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [RSC/N2612: Setup Extrusion Machine](#)
2. [RSC/N2613: Operate & Control Automated Rubber Extrusion Machine](#)
3. [RSC/N2614: Monitor Production and Perform Maintenance](#)
4. [RSC/N5007: Follow health & safety protocol](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Rubber
Sub-Sector	Industrial and Consumer Products
Occupation	industrial and consumer products production
Country	India
NSQF Level	4
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/8141.0300

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Minimum Educational Qualification & Experience	12th grade Pass (or Equivalent) OR 10th grade pass plus 2-year NTC (or NAC or CITS or equivalent) OR 10th grade pass with 3 Years of experience Relevant OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience Relevant
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	30/04/2028
NSQC Approval Date	30/04/2025
Version	4.0
Reference code on NQR	QG-04-RI-04034-2025-V2-RSDC
NQR Version	2

Qualification Pack

RSC/N2612: Setup Extrusion Machine

Description

This unit covers the preparation activities required before starting rubber extrusion: material verification, machine setup, die installation, temperature setting, and pre-operation safety checks.

Scope

The scope covers the following :

- Prepare work area and safety
- Select and inspect material
- Prepare extrusion machine
- Initial parameter setting and trial
- Documentation and handoff

Elements and Performance Criteria

Prepare work area and safety

To be competent, the user/individual on the job must be able to:

- PC1.** ensure work area is clean, well-lit, and hazard-free
- PC2.** place safety signage and remove unused items
- PC3.** wear PPE and check functionality
- PC4.** verify emergency stops and interlocks

Select and inspect material

To be competent, the user/individual on the job must be able to:

- PC5.** verify rubber compound type against job sheet
- PC6.** inspect compound for contamination or moisture
- PC7.** document material lot number and expiry
- PC8.** weigh compound accurately

Prepare extrusion machine

To be competent, the user/individual on the job must be able to:

- PC9.** clean barrel and screw before loading
- PC10.** fit correct die geometry and tightness
- PC11.** install hopper and feeding system
- PC12.** check heating zones and thermocouple connectivity
- PC13.** set temperature zones, screw speed, and feed rate
- PC14.** conduct trial run and adjust parameters
- PC15.** ensure consistent flow and correct shape
- PC16.** fine-tune cooling or cutter positioning

Documentation and handoff

To be competent, the user/individual on the job must be able to:

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- PC17.** record setup details and parameter values
- PC18.** communicate setup completion to incoming shift
- PC19.** respond to supervisor queries during setup
- PC20.** archive setup documentation per protocol

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** components of an extrusion line (barrel, screw, die, feeder)
- KU2.** properties of rubber compounds for extrusion (viscosity, cure)
- KU3.** die profile impact on product cross-section
- KU4.** heat zones and screw speed effects on melt quality
- KU5.** PPE and safety regulations in extrusion zones
- KU6.** machine pre-start checklist protocol
- KU7.** screw alignment and barrel inspection methods
- KU8.** procedure for material weighing and documentation
- KU9.** emergency and shutdown procedures
- KU10.** workplace housekeeping standards

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read job orders and machine manuals accurately
- GS2.** write clear setup logs and observations
- GS3.** communicate setup readiness with team members
- GS4.** apply arithmetic for weight and speed calculations
- GS5.** practice precision in manual adjustments

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare work area and safety</i>	8	12	-	-
PC1. ensure work area is clean, well-lit, and hazard-free	2	3	-	-
PC2. place safety signage and remove unused items	2	3	-	-
PC3. wear PPE and check functionality	2	3	-	-
PC4. verify emergency stops and interlocks	2	3	-	-
<i>Select and inspect material</i>	8	12	-	-
PC5. verify rubber compound type against job sheet	2	3	-	-
PC6. inspect compound for contamination or moisture	2	3	-	-
PC7. document material lot number and expiry	2	3	-	-
PC8. weigh compound accurately	2	4	-	-
<i>Prepare extrusion machine</i>	16	24	-	-
PC9. clean barrel and screw before loading	2	3	-	-
PC10. fit correct die geometry and tightness	2	3	-	-
PC11. install hopper and feeding system	2	3	-	-
PC12. check heating zones and thermocouple connectivity	2	3	-	-
PC13. set temperature zones, screw speed, and feed rate	2	3	-	-
PC14. conduct trial run and adjust parameters	2	3	-	-
PC15. ensure consistent flow and correct shape	2	3	-	-
PC16. fine-tune cooling or cutter positioning	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Documentation and handoff</i>	8	12	-	-
PC17. record setup details and parameter values	2	3	-	-
PC18. communicate setup completion to incoming shift	2	3	-	-
PC19. respond to supervisor queries during setup	2	3	-	-
PC20. archive setup documentation per protocol	2	3	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N2612
NOS Name	Setup Extrusion Machine
Sector	Rubber
Sub-Sector	Tyre and Non-tyre
Occupation	industrial and consumer products production
NSQF Level	4
Credits	3
Version	2.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

RSC/N2613: Operate & Control Automated Rubber Extrusion Machine

Description

This unit teaches safe, efficient operation of automated extrusion machinery, including continuous monitoring of output, making parameter adjustments, quality inspection, and coordinating with auxiliary systems.

Scope

The scope covers the following :

- Continuous monitoring
- Quality inspection
- Process control and adjustment
- Coordination with teams and compliance
- Emergency handling

Elements and Performance Criteria

Continuous monitoring

To be competent, the user/individual on the job must be able to:

- PC1.** monitor melt pressure, torque, and temperature zones
- PC2.** review HMI alarms and resolve minor alerts
- PC3.** check conveyor alignment and torque load
- PC4.** verify cooling water flow and temperature
- PC5.** oversee feeder alignment and material feeding

Quality inspection

To be competent, the user/individual on the job must be able to:

- PC6.** measure profile dimensions using calipers
- PC7.** inspect surface finish and color uniformity
- PC8.** check hardness or density if specified
- PC9.** identify defects (blisters, voids, flash)
- PC10.** segregate off-spec products

Process control and adjustment

To be competent, the user/individual on the job must be able to:

- PC11.** adjust speed, temperature, pressure for consistency
- PC12.** fine-tune die height and cutter timing
- PC13.** optimize feeder pressure for compound flow
- PC14.** coordinate with auxiliary cooling and winding

Coordination with teams and compliance

To be competent, the user/individual on the job must be able to:

- PC15.** liaise with QA for product sampling

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- PC16.** coordinate with production planners for batch targets
- PC17.** communicate deviations during shift handover
- PC18.** notify maintenance for unresolved mechanical issues
- PC19.** maintain extrusion logs and reject reports
- PC20.** document parameter changes and time stamps
- PC21.** update digital production reports
- PC22.** validate calibration status of sensors

Emergency handling

To be competent, the user/individual on the job must be able to:

- PC23.** initiate emergency stop procedures as needed
- PC24.** document incident and corrective actions
- PC25.** participate in basic incident investigation

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** HMI/SCADA system principles and navigation
- KU2.** process parameters and their effect on product quality
- KU3.** common extrusion defects and troubleshooting methods
- KU4.** quality standards for rubber extrudates
- KU5.** sensor calibration and validation techniques
- KU6.** lockout-tagout and emergency procedures
- KU7.** interrelation of auxiliary systems in production
- KU8.** data recording and digital reporting standards
- KU9.** root cause analysis basics for extrusion faults
- KU10.** shift handover protocols and teamwork importance
- KU11.** waste reduction and process improvement strategies
- KU12.** ergonomics and safety during operations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** interpret digital displays and trend graphs
- GS2.** report production metrics using digital tools
- GS3.** collaborate with QA and maintenance quickly
- GS4.** apply calculation skills for yield and efficiency
- GS5.** resolve process alerts proactively

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Continuous monitoring</i>	10	15	-	-
PC1. monitor melt pressure, torque, and temperature zones	2	3	-	-
PC2. review HMI alarms and resolve minor alerts	2	3	-	-
PC3. check conveyor alignment and torque load	2	3	-	-
PC4. verify cooling water flow and temperature	2	3	-	-
PC5. oversee feeder alignment and material feeding	2	3	-	-
<i>Quality inspection</i>	10	14	-	-
PC6. measure profile dimensions using calipers	2	3	-	-
PC7. inspect surface finish and color uniformity	2	3	-	-
PC8. check hardness or density if specified	2	3	-	-
PC9. identify defects (blisters, voids, flash)	2	3	-	-
PC10. segregate off-spec products	2	2	-	-
<i>Process control and adjustment</i>	8	11	-	-
PC11. adjust speed, temperature, pressure for consistency	2	3	-	-
PC12. fine-tune die height and cutter timing	2	3	-	-
PC13. optimize feeder pressure for compound flow	2	3	-	-
PC14. coordinate with auxiliary cooling and winding	2	2	-	-
<i>Coordination with teams and compliance</i>	9	14	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. liaise with QA for product sampling	2	2	-	-
PC16. coordinate with production planners for batch targets	1	2	-	-
PC17. communicate deviations during shift handover	1	2	-	-
PC18. notify maintenance for unresolved mechanical issues	1	1	-	-
PC19. maintain extrusion logs and reject reports	1	2	-	-
PC20. document parameter changes and time stamps	1	1	-	-
PC21. update digital production reports	1	2	-	-
PC22. validate calibration status of sensors	1	2	-	-
<i>Emergency handling</i>	3	6	-	-
PC23. initiate emergency stop procedures as needed	1	2	-	-
PC24. document incident and corrective actions	1	2	-	-
PC25. participate in basic incident investigation	1	2	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N2613
NOS Name	Operate & Control Automated Rubber Extrusion Machine
Sector	Rubber
Sub-Sector	Tyre and Non-tyre
Occupation	industrial and consumer products production
NSQF Level	4
Credits	6
Version	2.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

RSC/N2614: Monitor Production and Perform Maintenance

Description

This unit equips candidates with skills to monitor continuous extrusion operations, perform routine maintenance, detect performance decline, and execute minor troubleshooting or machine care tasks.

Scope

The scope covers the following :

- Routine monitoring
- Preventive maintenance
- Calibration & sensor care
- Troubleshooting & minor repairs
- Waste disposal & Team Coordination

Elements and Performance Criteria

Routine monitoring

To be competent, the user/individual on the job must be able to:

- PC1.** check line output vs target speeds
- PC2.** record temperature and pressure values periodically
- PC3.** inspect cooling and cutting system function
- PC4.** log any variation from normal process trends

Preventive Maintenance

To be competent, the user/individual on the job must be able to:

- PC5.** schedule cleaning for die, barrel, and screw
- PC6.** lubricate bearings, gearbox, and feeder units
- PC7.** replace filters, seals, and worn linings
- PC8.** tighten bolts per torque specifications

Calibration & sensor care

To be competent, the user/individual on the job must be able to:

- PC9.** calibrate temperature and pressure sensors
- PC10.** verify calibration with standard references
- PC11.** document calibration dates and results

Troubleshooting & minor repairs

To be competent, the user/individual on the job must be able to:

- PC12.** inspect mechanical components for wear
- PC13.** replace gaskets, belts, and screws
- PC14.** adjust feeder tension or alignment
- PC15.** debug minor electrical faults under guidance

Waste disposal and team coordination

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To be competent, the user/individual on the job must be able to:

- PC16.** dispose scrap rubber and waste fluid safely
- PC17.** clean tooling and storage areas
- PC18.** manage spare parts inventory and reorder
- PC19.** update maintenance and safety logs
- PC20.** communicate wear signs to maintenance team
- PC21.** assist during preventive shutdowns
- PC22.** support shift handover updates

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** wear mechanisms in extrusion screws and dies
- KU2.** preventive maintenance principles and schedules
- KU3.** sensor function and calibration procedures
- KU4.** lockout-tagout and maintenance safety
- KU5.** common mechanical faults and diagnostic steps
- KU6.** lubricant and grease types and application
- KU7.** waste categorization and disposal guidelines
- KU8.** inventory management and reorder levels
- KU9.** record-keeping techniques for maintenance logs
- KU10.** impact of maintenance on product quality and uptime
- KU11.** communication standards during shutdowns and alerts

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and update maintenance work orders and logs
- GS2.** write simple maintenance reports and summaries
- GS3.** coordinate with production shifts about shutdowns
- GS4.** apply measurement tools for fault diagnosis
- GS5.** schedule and track preventive maintenance tasks

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Routine monitoring</i>	7	9	-	-
PC1. check line output vs target speeds	1	2	-	-
PC2. record temperature and pressure values periodically	2	3	-	-
PC3. inspect cooling and cutting system function	2	3	-	-
PC4. log any variation from normal process trends	2	1	-	-
<i>Preventive Maintenance</i>	8	12	-	-
PC5. schedule cleaning for die, barrel, and screw	2	3	-	-
PC6. lubricate bearings, gearbox, and feeder units	2	3	-	-
PC7. replace filters, seals, and worn linings	2	3	-	-
PC8. tighten bolts per torque specifications	2	3	-	-
<i>Calibration & sensor care</i>	6	9	-	-
PC9. calibrate temperature and pressure sensors	2	3	-	-
PC10. verify calibration with standard references	2	3	-	-
PC11. document calibration dates and results	2	3	-	-
<i>Troubleshooting & minor repairs</i>	8	12	-	-
PC12. inspect mechanical components for wear	2	3	-	-
PC13. replace gaskets, belts, and screws	2	3	-	-
PC14. adjust feeder tension or alignment	2	3	-	-
PC15. debug minor electrical faults under guidance	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Waste disposal and team coordination</i>	11	18	-	-
PC16. dispose scrap rubber and waste fluid safely	2	3	-	-
PC17. clean tooling and storage areas	2	3	-	-
PC18. manage spare parts inventory and reorder	2	3	-	-
PC19. update maintenance and safety logs	2	3	-	-
PC20. communicate wear signs to maintenance team	1	2	-	-
PC21. assist during preventive shutdowns	1	2	-	-
PC22. support shift handover updates	1	2	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N2614
NOS Name	Monitor Production and Perform Maintenance
Sector	Rubber
Sub-Sector	Tyre and Non-tyre
Occupation	industrial and consumer products production
NSQF Level	4
Credits	4
Version	2.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

RSC/N5007: Follow health & safety protocol

Description

This unit is about maintaining health and safety of self and others at workplace.

Scope

The scope covers the following :

- Maintain a clean and efficient workplace
- Render appropriate emergency procedures
- Maintain standard safety procedures at the workplace
- Participate in safety awareness campaigns
- Understand potential sources of accidents
- Use safety gears to avoid accidents

Elements and Performance Criteria

Maintain a clean and efficient workplace

To be competent, the user/individual on the job must be able to:

- PC1.** undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor
- PC2.** work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy
- PC3.** read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc
- PC4.** prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.
- PC5.** use equipment and materials safely and correctly and return the same to designated storage when not in use
- PC6.** dispose off waste safely and correctly in a designated area
- PC7.** risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace
- PC8.** perform work in a manner which minimizes environmental damage
- PC9.** all procedures and work instructions for controlling risk are followed closely
- PC10.** report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger

Render appropriate emergency procedures

To be competent, the user/individual on the job must be able to:

- PC11.** follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.
- PC12.** follow emergency procedures as per company standards and workplace requirements

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- PC13.** use emergency equipment in accordance with manufacturers' specifications and workplace requirements
- PC14.** provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.
- PC15.** recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate
- PC16.** dispose off medical waste in accordance with workplace requirements
- PC17.** report details of first aid administered in accordance with work place procedures

Maintain standard safety procedures at the workplace

To be competent, the user/individual on the job must be able to:

- PC18.** comply with general safety procedures
- PC19.** follow standard safety procedures while handling equipment, hazardous material or tool
- PC20.** check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.
- PC21.** ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure
- PC22.** keep the workplace organized, swept, clean and hazard free

Participate in safety awareness campaigns

To be competent, the user/individual on the job must be able to:

- PC23.** attend fire drills and other safety related workshops organized at the workplace
- PC24.** be aware of first aid, evacuation and emergency procedures
- PC25.** be alert of any events and do not be negligent to any safety procedures to be followed

Understand potential sources of accidents

To be competent, the user/individual on the job must be able to:

- PC26.** avoid accidents while using hazardous chemicals, machines, sharp tools and equipment

Use safety gears to avoid accidents

To be competent, the user/individual on the job must be able to:

- PC27.** use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)
- PC28.** handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies on incentives, delivery standards, and personnel management.
- KU2.** occupational safety and health policy followed
- KU3.** emergency evacuation procedure
- KU4.** medical policy
- KU5.** company laws and acts
- KU6.** the risks to health and safety and the measures to be taken to control those risks in the area of work

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- KU7.** workplace procedures and requirements for the handling of workplace injuries/illnesses.
- KU8.** basic emergency first aid procedure
- KU9.** local emergency services
- KU10.** reporting on accidents, incidents and problems to appropriate authorities.
- KU11.** how to use machines as per standard operating procedure
- KU12.** how to maintain work area safe and secure
- KU13.** use of hazardous materials, tools and equipments
- KU14.** emergency evacuation and first aid procedures to be followed
- KU15.** personal hygiene and fitness requirements
- KU16.** general duties under the relevant health and safety legislation
- KU17.** what personal protective equipment and clothing should be worn and how it is cared for
- KU18.** the correct and safe way to use materials and equipment required for work
- KU19.** the importance of good housekeeping in the workplace
- KU20.** safe disposal methods for waste
- KU21.** methods for minimizing environmental damage during work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record data which are required for record keeping purpose
- GS2.** report problems to the appropriate person in a timely manner
- GS3.** write descriptions and details about incidents in reports
- GS4.** read instruction manuals for hand tools and equipment
- GS5.** read instructions on work orders and procedures
- GS6.** receive instructions and seek advice from superiors
- GS7.** communicate clearly and effectively with others
- GS8.** take a decision for any change/issue based on earlier successes(documented previous history)on similar issues
- GS9.** work out changes in case a new improved machine/equipment is added in the process or any new material/chemical is developed replacing existing one.
- GS10.** make changes in cycle time due to improved process.
- GS11.** use the standard operating procedure or trouble shooting manuals for trouble shooting and other reference documents approved by plant management
- GS12.** consult the peer group and superiors to arrive at a favourable decision.
- GS13.** use of standard available problem solving techniques for decision making
- GS14.** review and analyze the process steps to check on system non adherence andnon conformity
- GS15.** review the current sop and other standards for continuous improvement to facilitate decision making
- GS16.** take a calculated risk with minimum losses
- GS17.** schedule daily activities and drawing up priorities; allocate start times, estimation of completion times and materials, equipment and assistance required for completion.

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- GS18.** match customer needs/specification by adjusting the processing conditions (interact with customer in case any clarification required)
- GS19.** ensure that performance of his action/operation/activity does not lead to any divergence from the specified quality of the final product as required by the customer.
- GS20.** complete the assigned task in timely manner so that the final product is delivered in the timeline given by the customer.
- GS21.** communicate effectively to the superior/customer for any delay in supplies to the clients.
- GS22.** work towards fulfilling the customers requirement as per their demand.
- GS23.** in case of any complaint, ensure its timely resolution if the problem is emanating at his level
- GS24.** communicate effectively to the superior/customer for any delay in resolving the problem faced by the customer.
- GS25.** maintain good/cordial relation with customers.
- GS26.** work on the feedback received from customer regarding the product.
- GS27.** use first aid treatment in case of any injury/accident.
- GS28.** monitor and maintain the condition of tools and equipment
- GS29.** assess situation & identify appropriate control measures
- GS30.** act, communicate and report in emergency situation

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain a clean and efficient workplace</i>	9	21	-	-
PC1. undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor	2	2	-	-
PC2. work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy	2	2	-	-
PC3. read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc	-	2	-	-
PC4. prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.	2	2	-	-
PC5. use equipment and materials safely and correctly and return the same to designated storage when not in use	1	2	-	-
PC6. dispose off waste safely and correctly in a designated area	2	4	-	-
PC7. risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace	-	2	-	-
PC8. perform work in a manner which minimizes environmental damage	-	2	-	-
PC9. all procedures and work instructions for controlling risk are followed closely	-	1	-	-
PC10. report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger	-	2	-	-
<i>Render appropriate emergency procedures</i>	9	18	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.	2	4	-	-
PC12. follow emergency procedures as per company standards and workplace requirements	2	4	-	-
PC13. use emergency equipment in accordance with manufacturers' specifications and workplace requirements	2	4	-	-
PC14. provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.	-	1	-	-
PC15. recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate	-	2	-	-
PC16. dispose off medical waste in accordance with workplace requirements	-	1	-	-
PC17. report details of first aid administered in accordance with work place procedures	3	2	-	-
<i>Maintain standard safety procedures at the workplace</i>	6	15	-	-
PC18. comply with general safety procedures	2	4	-	-
PC19. follow standard safety procedures while handling equipment, hazardous material or tool	-	2	-	-
PC20. check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.	2	4	-	-
PC21. ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure	-	1	-	-
PC22. keep the workplace organized, swept, clean and hazard free	2	4	-	-
<i>Participate in safety awareness campaigns</i>	2	8	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. attend fire drills and other safety related workshops organized at the workplace	1	3	-	-
PC24. be aware of first aid, evacuation and emergency procedures	1	3	-	-
PC25. be alert of any events and do not be negligent to any safety procedures to be followed	-	2	-	-
<i>Understand potential sources of accidents</i>	1	3	-	-
PC26. avoid accidents while using hazardous chemicals, machines, sharp tools and equipment	1	3	-	-
<i>Use safety gears to avoid accidents</i>	3	5	-	-
PC27. use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)	2	2	-	-
PC28. handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders	1	3	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N5007
NOS Name	Follow health & safety protocol
Sector	Rubber
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4
Credits	TBD
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
RSC/N2612.Setup Extrusion Machine	40	60	-	-	100	15
RSC/N2613.Operate & Control Automated Rubber Extrusion Machine	40	60	-	-	100	40
RSC/N2614.Monitor Production and Perform Maintenance	40	60	-	-	100	25
RSC/N5007.Follow health & safety protocol	30	70	-	-	100	10
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	170	280	-	-	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.